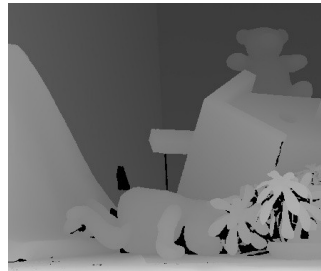


Stereo Vision

- Compute depth from two views
- Many things left unsaid: what is known, and what is unknown



Disparity Map

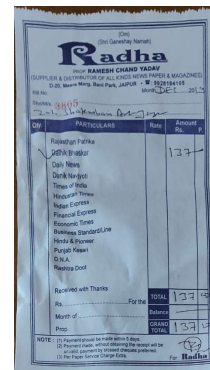
1

Single View

Nothing ==
casual picture

- Given a single view picture
 - If we know “nothing” about the camera
- We can still do useful things

Nothing: the camera is uncalibrated, i.e., we know neither the intrinsic, nor the extrinsic parameters



2

Single View

- Given a picture,
 - If we know nothing about the camera,
 - We know *something* about the scene
- We can do useful things: single view metrology

Something: Two known distances

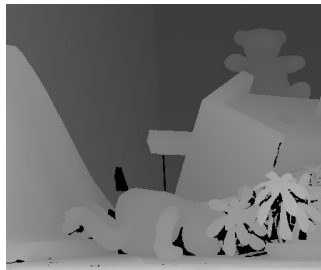


3

Stereo Vision

Nothing ==
casual picture

- Compute depth from two views
- When we are given **casual pictures**, not necessarily from a device such as a Kinect or *calibrated* cameras



4

Overview

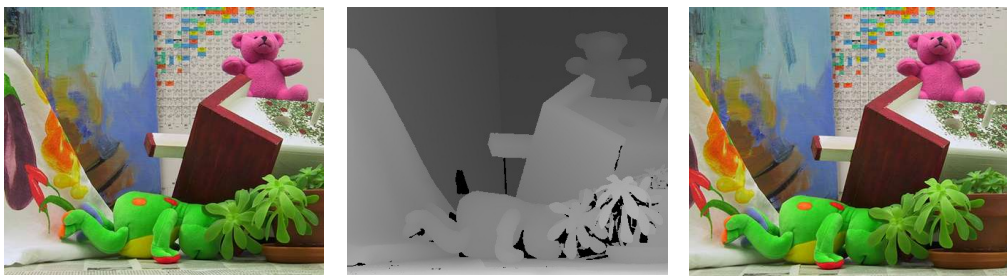
- How to do reconstruction if 'parameters' are known
- Some fun stuff!!
- Basics, buzzwords, and terminology on stereo
- The fundamental matrix F
 - Computing 'parameters' from F
- The essential matrix E
 - Computing 'parameters' from E

5

Stereo Vision: Known cameras

- Compute depth from two views

Not Casual: Stereo parameters (R, T) are known with 100% accuracy



6

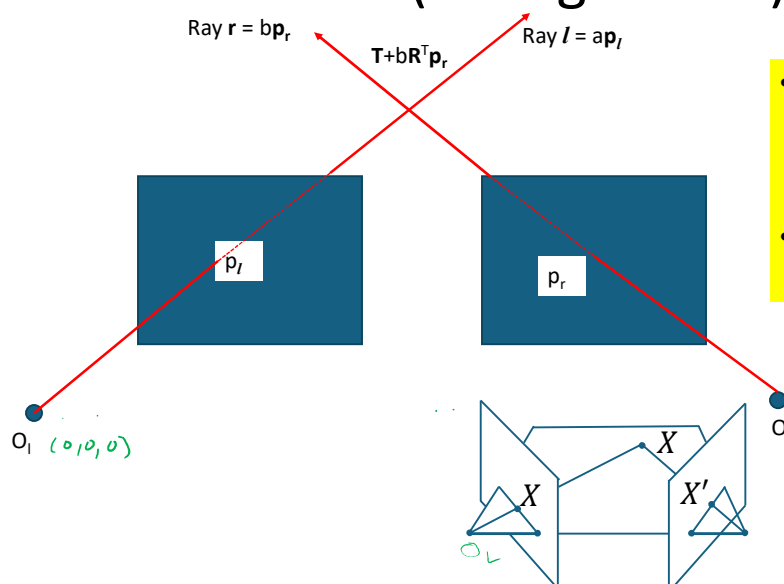
Parameters of a stereo system

- Intrinsic parameters – focal lengths, optical centers, camera resolutions (may be same)
 - Transformations mapping a 3D point to pixel coordinates in each camera
- Extrinsic parameters – rotation R and translation T that align the coordinate systems of the two cameras
 - Transformations that align the relative position and orientation of the two cameras

$$X_{3 \times 1} = K_{3 \times 3} [I_{3 \times 3} \mid 0] X_{\text{cam}} \quad K = \begin{bmatrix} f & & p_x \\ & f & p_y \\ & & 1 \end{bmatrix}$$

7

3D Reconstruction (Triangulation)



$$\vec{a} \vec{p}_l \quad a \in \mathbb{R}$$

$$K^T P_L \cdot (23, 45)$$

- Stereo parameters (R, T) are known with 100% accuracy
- Cameras are in general position

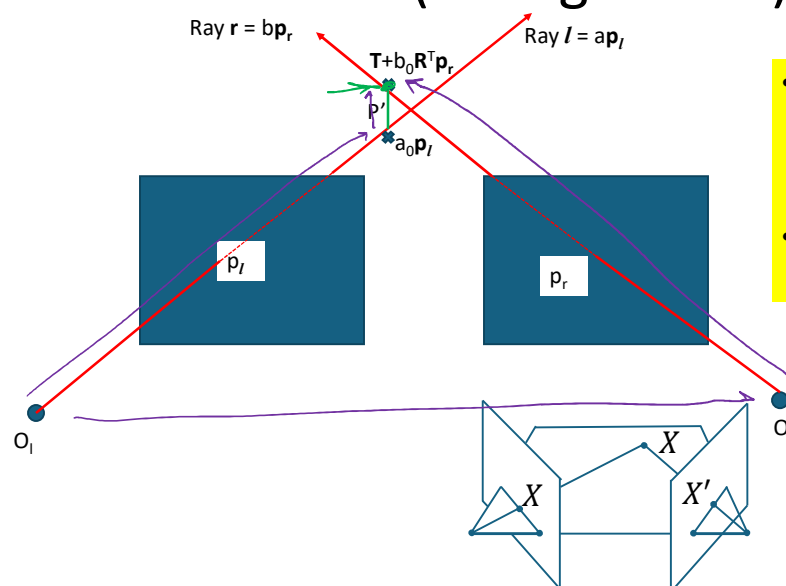
8

3D reconstruction: Known parameters

- Rays r and l may not intersect in practice
 - For example, due to measurement errors
- Therefore, find shortest line segment s between two rays

9

3D Reconstruction (Triangulation)



- Stereo parameters (R, T) are known with 100% accuracy
- Cameras are in general position

10

3D reconstruction: Known parameters

- Rays $\mathbf{r}$ and $\mathbf{l}$ may not intersect in practice
 - For example, due to measurement errors
- Therefore, find shortest line segment $\mathbf{s}$ between two rays
 - Line segment $\mathbf{s}$ perpendicular to both $\mathbf{r}$ and $\mathbf{l}$
- Thus, we have $\mathbf{s}$ lying in the direction of vector $\mathbf{w} = \mathbf{p}_l \times \mathbf{R}^T \mathbf{p}_r$
 - One end on $\mathbf{l}$ and another on $\mathbf{r}$
 - Treat midpoint of $\mathbf{s}$ as the point of intersection
 - The midpoint is the point of minimum distance from both $\mathbf{r}$ and $\mathbf{l}$

11

3D reconstruction: Known parameters → $a_0 \mathbf{p}_l$

- The concerned segment starts at point $a_0 \mathbf{p}_l$ on ray $\mathbf{l}$ and ends at point $\mathbf{T} + b_0 \mathbf{R}^T \mathbf{p}_r$ on ray $\mathbf{r}$
- Noting that segment $\mathbf{s}$ lies in the direction of ray $\mathbf{w}$, a point on segment $\mathbf{s}$ can be expressed as $a_0 \mathbf{p}_l + c_0 \mathbf{w}$
 - $a_0 \mathbf{p}_l + c_0 \mathbf{w} = a_0 \mathbf{p}_l + c_0 (\mathbf{p}_l \times \mathbf{R}^T \mathbf{p}_r)$
- Hence, we have $\mathbf{T} + b_0 \mathbf{R}^T \mathbf{p}_r = a_0 \mathbf{p}_l + c_0 (\mathbf{p}_l \times \mathbf{R}^T \mathbf{p}_r)$
 - Solve for coefficients a_0, b_0, c_0

$$\begin{aligned}
 & \text{Handwritten notes and diagram:} \\
 & \text{Diagram shows a triangle with vertices } a_0 \mathbf{p}_l, \mathbf{T} + b_0 \mathbf{R}^T \mathbf{p}_r, \text{ and } a_0 \mathbf{p}_l + c_0 \mathbf{w}. \\
 & \text{Equation: } \mathbf{T} + b_0 \mathbf{R}^T \mathbf{p}_r = a_0 \mathbf{p}_l + c_0 (\mathbf{p}_l \times \mathbf{R}^T \mathbf{p}_r) \\
 & \text{Component-wise equations:} \\
 & \quad t_x + b_0 a_x = a_0 d_x + c_0 w_x \\
 & \quad t_y + b_0 a_y = a_0 d_y + c_0 w_y \\
 & \quad t_z + b_0 a_z = a_0 d_z + c_0 w_z
 \end{aligned}$$

12

3D reconstruction: Known parameters Handwritten: $O \rightarrow O'$

- Moral of the story (is stereo hard?)
 - With known camera parameters,
 - 3D reconstruction is essentially unambiguous.
 - Accuracy depends on noise level.

Stereo parameters (R, T) are known with 100% accuracy

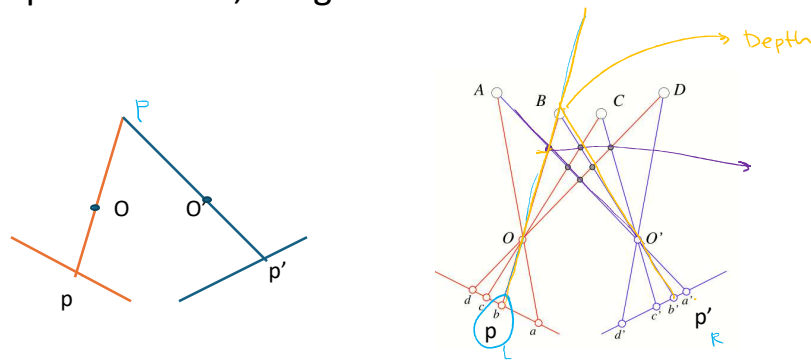
Key challenge remains

13

Key Challenge

Correspondence Vision

- Given “casual” left and right images, how can we (efficiently) find the correspondence for random points in the scene?
 - 2D example is chosen, imagine the situation in 3D



14

3D reconstruction: Known parameters

- Moral of the story (is stereo hard?)
 - With known camera parameters,
 - Known correspondence modulo noise
 - 3D reconstruction is essentially unambiguous.
 - Accuracy depends on noise level.

Stereo parameters (R, T) are known with 100% accuracy

Correspondence is known

15

3D reconstruction with Stereo

- Moral of the story (is stereo hard?)
 - Geometry
 - Correspondence

16

Overview

- How to do reconstruction if parameters are known
- Some **fun** stuff!!
- Basics, buzzwords, and terminology on stereo
- The fundamental matrix $\mathbf{F}$
 - Computing 'parameters' from $\mathbf{F}$
- The essential matrix $\mathbf{E}$
 - Computing 'parameters' from $\mathbf{E}$

17

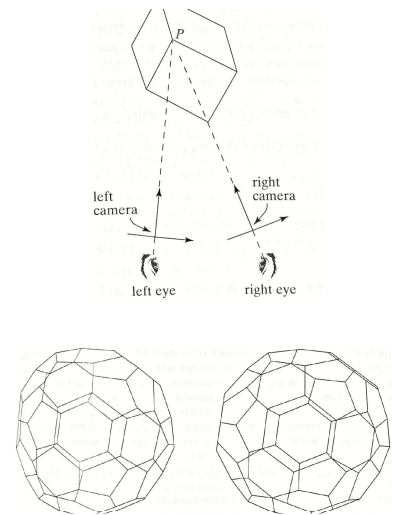
Stereo Vision

- First a message from your sponsor
 - Hold your index finger an arm's length away
 - Look at it through the left eye (keep right eye closed)
 - Now look at it through the right eye (keep left closed)
 - You will perceive a shift - this is called stereo disparity and
- The brain can use this significantly to infer depth
 - although it is not necessary

18

Stereo Vision

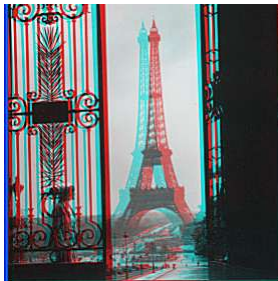
- CG: generate two pictures of the (same) scene from a slightly different viewpoint
- In this picture it's possible to coax the eyes to see a 3D view



19

Stereo Vision

- Can use color channels and prevent one of the eyes from seeing the "other"



20

Review

- How to do reconstruction if parameters are known
 - Unaligned cameras
- Some fun stuff (using our index finger)
 - Parallax
 - Was there any vertical shift?

21

Overview

- Some fun stuff!!
- How to do reconstruction if parameters are known
 - Unaligned cameras
 - *Aligned cameras*

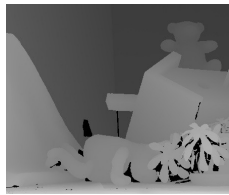
22

Stereo Vision

- Compute depth from two views
 - ~~When we are given casual pictures, not necessarily from a device such as a Kinect or calibrated cameras~~

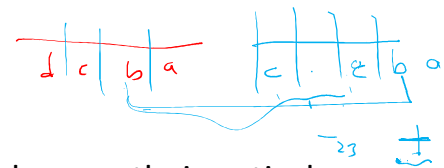
Not given casual pictures

In the special case, the pictures are far from casual; very specific cameras

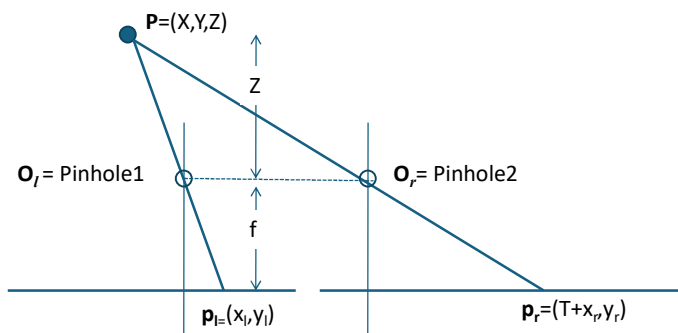


23

Simplest Case



- Assume pinhole positions of two cameras are known, their optical axes are aligned (parallel)
 - Want to compute depth (done this earlier with unaligned cameras!)



- Line $O_l O_r$ = **baseline**.
- Assume baseline is perpendicular to **both** optical axes. Camera X-axis is parallel to the baseline.
- Let length $O_l O_r = T$ = baseline length. Focal length = f .
- Triangles $PO_l O_r$ and $Pp_l p_r$ are similar.

24

Simplest Case

- From similarity of triangles:

$$\frac{T}{Z} = \frac{T + x_r - x_l}{Z + f}$$

$$Z = \frac{fT}{x_r - x_l} = \frac{fT}{d}$$

$$y_l = y_r = f \frac{Y}{Z}$$

Disparity – a spatially varying quantity. At each point (x, y) in the left image, we have disparity $d(x, y)$ and $x + d(x, y)$ is the x-coordinate of its corresponding point in the second image.

Note that y-coordinates are equal because of our assumptions: X axis of the cameras coincides with the baseline (the optical centers collapses to the same point) and the two imaging planes are co-planar

25

Disparity

- Disparity directly proportional to baseline length
 - Large baseline is a problem (due to missing correspondences as fields of view are different!)
 - Small baseline is error prone since Z is inversely to disparity

$$Z = \frac{fT}{x_r - x_l} = \frac{fT}{d}$$

$$\frac{\partial Z}{\partial d} = -\frac{fT}{d^2}$$

What conclusions can we draw **only** from this simplest case?

26

Disparity

- Technically disparity increases as the distance from the fixation point (intersection of the optical axes, ∞ in the present case)
 - Distance to faraway objects can be measured less accurately than to nearby ones

$$Z = \frac{fT}{x_r - x_l} = \frac{fT}{d}$$

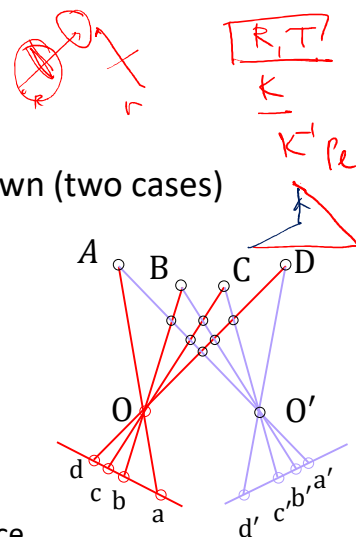
$$\frac{\partial Z}{\partial d} = -\frac{fT}{d^2}$$

What conclusions can we draw **only** from this simplest case?

27

Review

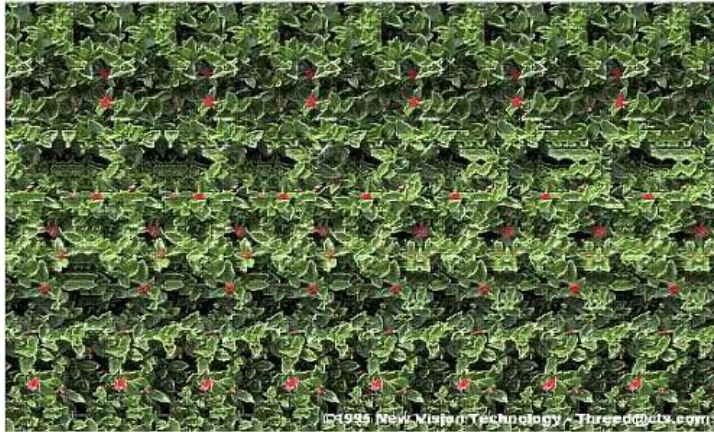
- How to do reconstruction if parameters are known (two cases)
 - Unaligned optical axis
 - Aligned optical axis
- Some fun stuff (using our index finger)
 - Parallax
- Buzzword
 - Disparity
 - Basics ("what's the main problem?") : Correspondence
- More fun stuff



29

Autostereograms

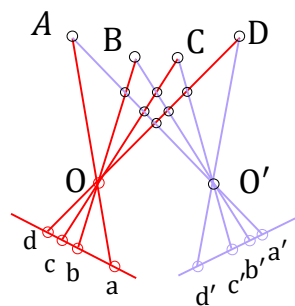
- Exploit disparity as depth cue using single image



30

Key Challenge

- Given “casual” left and right images, how can we (efficiently) find the correspondence for random points in the scene?
 - The “fundamental” matrix is an important step
 - (if we are confident on some 7 corresponding points)



31